L5 Scale Free Networks and Power Law Distribution

COM 5506
Social Network Analysis for Communication

SEM B 2025-2026

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Recapitulation

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- Random network model (Erdos-Renyi model):
  - a good starting point but too idealistic to exist in society
- Small world network (Watts-Strogatz model):
  - take into consideration clustering and average shortest path, similar to human social network
  - but ignored the issue of degree distribution – in other words, the models we looked at so far did not consider or care about the percentage of people receiving a particular value of degrees
- Today and next time: power law distribution scale free network and its properties

Distribution & Probability

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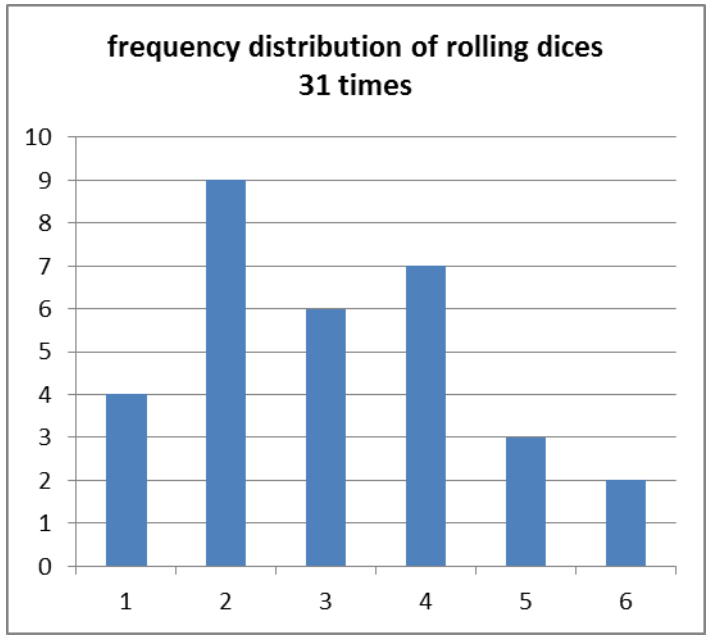
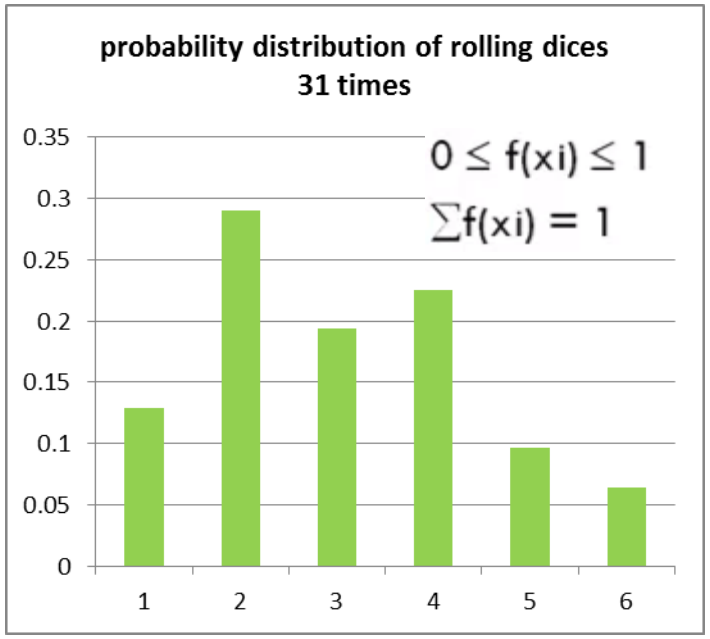
- Frequency distribution: a list of the values recorded in a sample
- Probability distribution: the probability of a particular value or value range of a variable

frequency distribution table

A data table that lists a set of scores and their frequency.

score	tally	frequency (f)
1	IIII	4
2	~~IIII~~ IIII	9
3	~~IIII~~ I	6
4	~~IIII~~ II	7
5	III	3
6	II	2





Distribution & Probability

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- Supposing we have a fair dice, what is the probability distribution of all the events?

outcome	1	2	3	4	5	6
expected value	1/6	1/6	1/6	1/6	1/6	1/6

Distribution & Probability

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- Supposing we have two fair dice, what is the probability distribution of all the events?

Events	2	3	4	5	6	7	8	9	10	11	12

Events	2	3	4	5	6	7	8	9	10	11	12
probability	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

Rolling 1 dice vs. 2 dices

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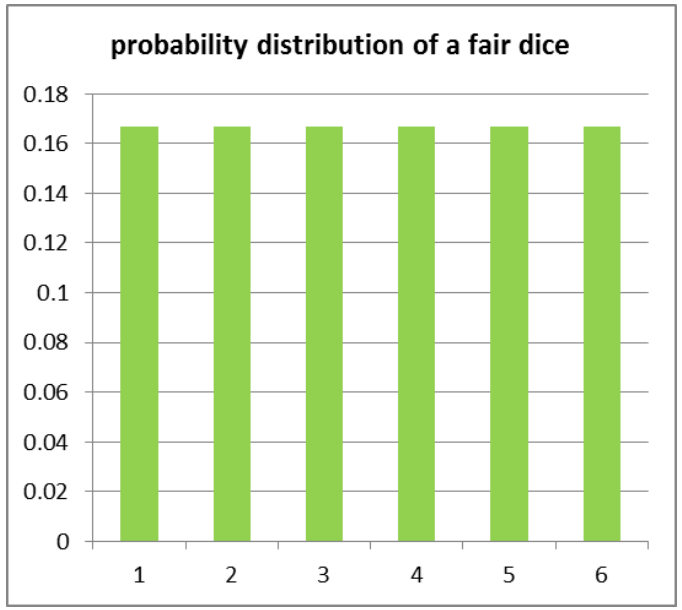

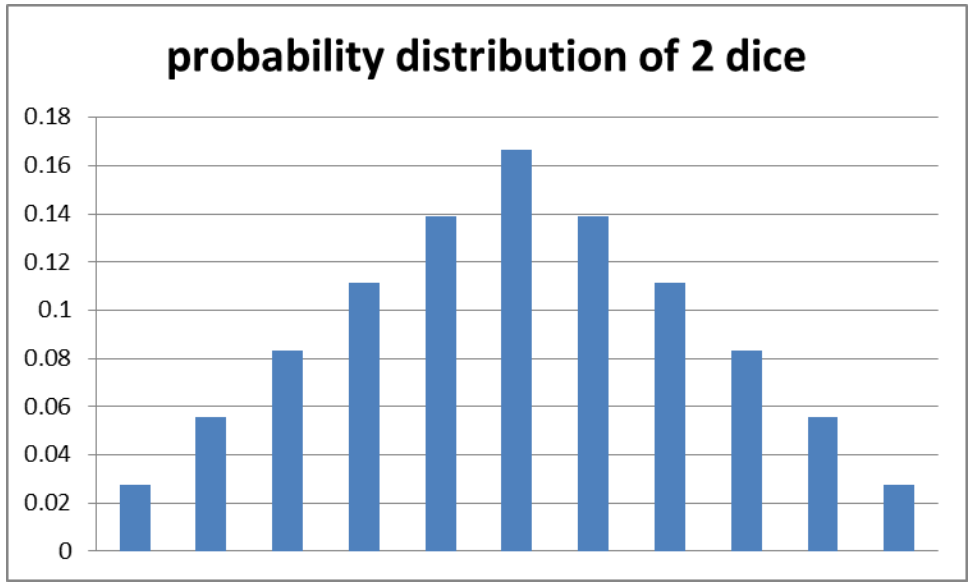

2 3 4 5 6 7 8 9 10 11 12

f(xi)=0.166666….

$$P(x) = \begin{cases} \frac{1}{36} & \text{if } x \in \{2, 12\} \\ \frac{2}{36} = \frac{1}{18} & \text{if } x \in \{3, 11\} \\ \frac{3}{36} = \frac{1}{12} & \text{if } x \in \{4, 10\} \\ \frac{4}{36} = \frac{1}{9} & \text{if } x \in \{5, 9\} \\ \frac{5}{36} & \text{if } x \in \{6, 8\} \\ \frac{6}{36} = \frac{1}{6} & \text{if } x = 7 \end{cases}$$

This is called density functions

Properties of probability density functions (pdf)

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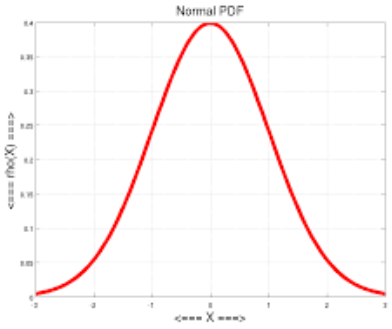
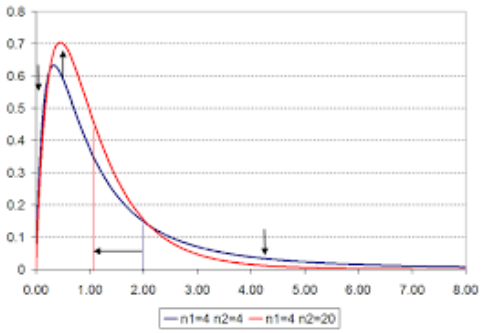
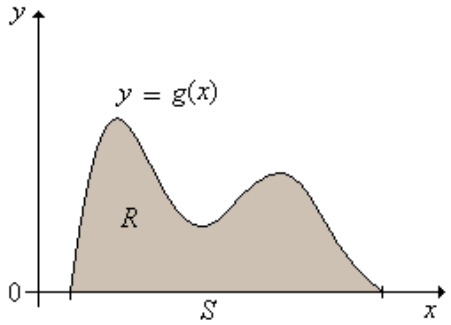
- f(x)>=0 for all x
- Total area under f(x)=1, in other words, the integral of f(x) within the domain of f(x) equals 1, i.e., $\int f(x) = 1$
- If f(x) is continuous rather than discrete, f(X=x)=0.
- For continuous function, we can only define probabilities over intervals: p(a<x<b), p(x<a), etc.







Cumulative distribution function (cdf)

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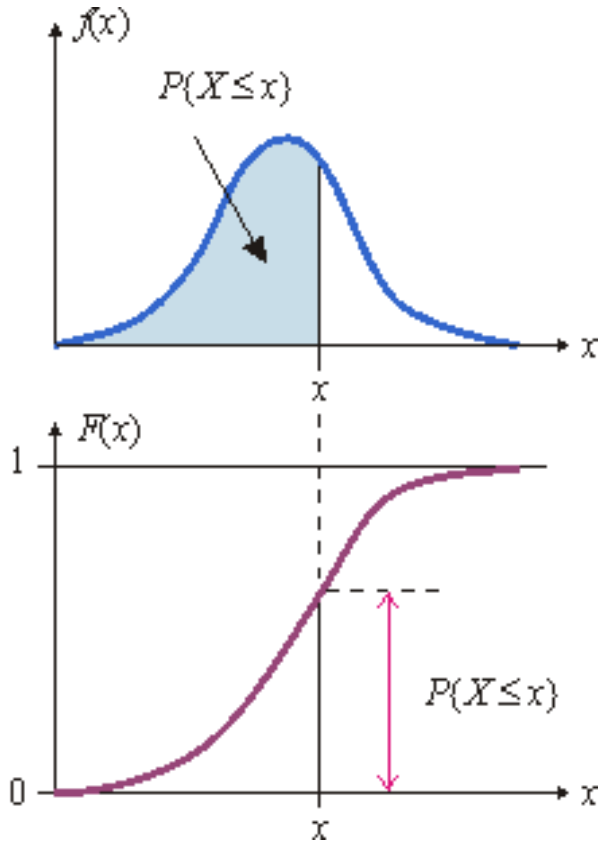
- f(x) is probability density function; F(x) refers to the probability that variable X will be less than or equal to x; that is, P(X>=x).
- F(x) is equal to the area under f(x) to the left of x
- The probability that X is between a and b is the area under f(x) from a to b=F(b)-F(a)



Cumulative distribution function (cdf)

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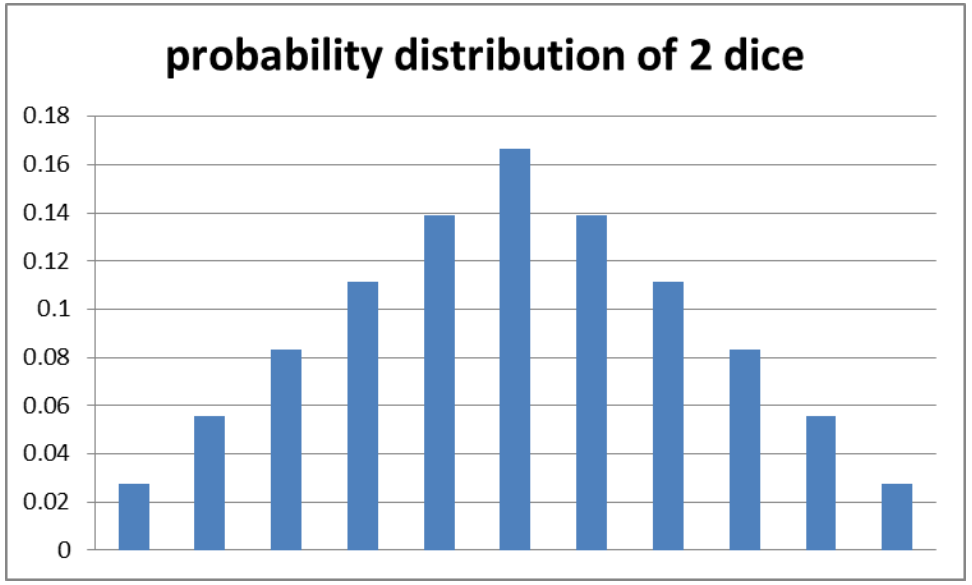
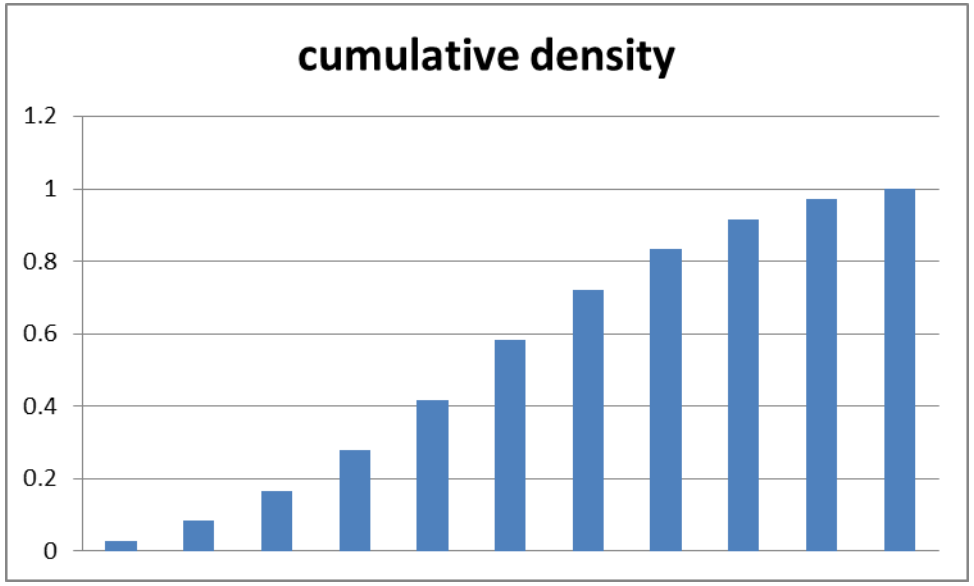
Events	2	3	4	5	6	7	8	9	10	11	12
probability	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36
cumulative p.											



2 3 4 5 6 7 8 9 10 11 12



2 3 4 5 6 7 8 9 10 11 12

What is
the probability
between 6 and 9?

Gaussian distribution
Normal distribution

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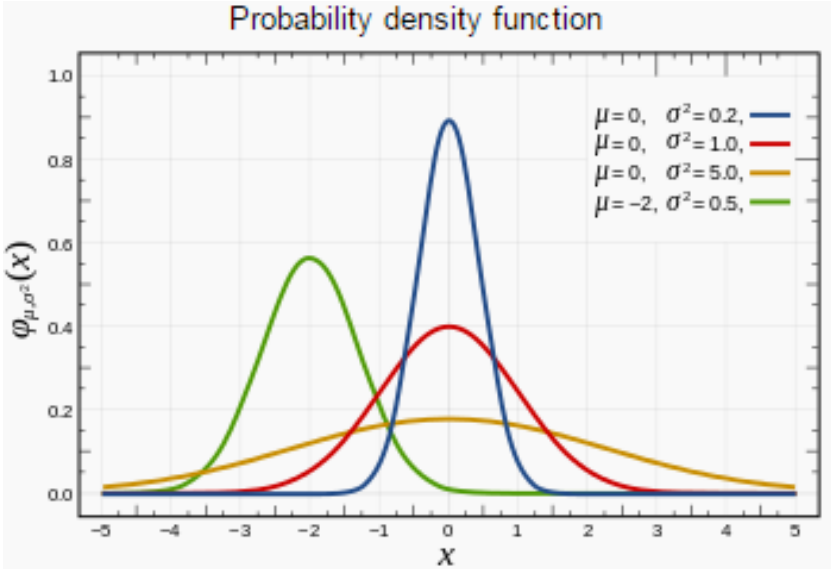
- Bell-shape curve
- Symmetric, median=mean=mode
- Range unbounded
- Mean μ, variance σ^2
- Density function

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\sigma^2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- The result of a random process
- height, weight, intelligence, length of sleep, but NOT necessarily wealth…



The red curve is the *standard normal distribution*

Gaussian distribution examples

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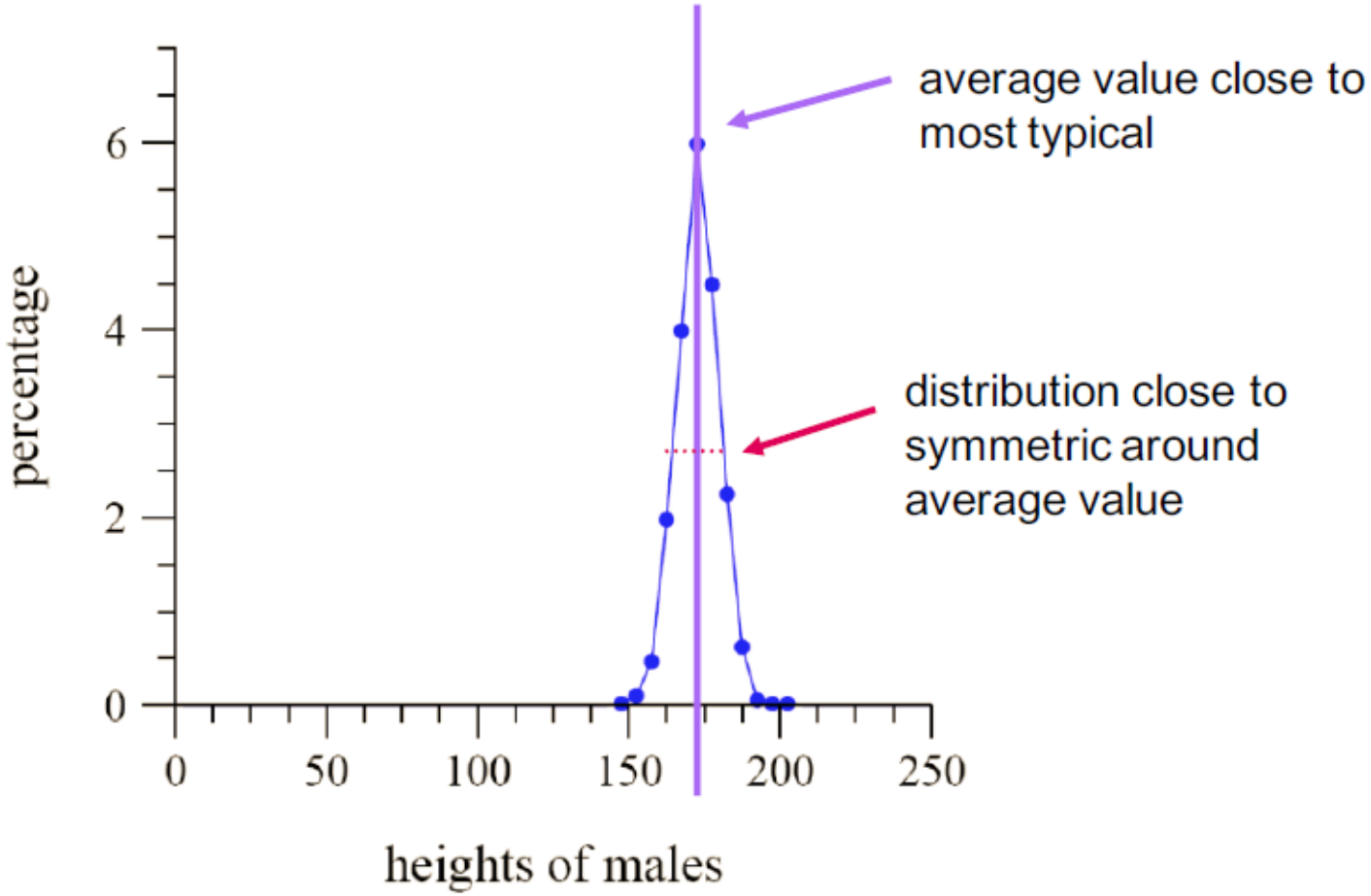

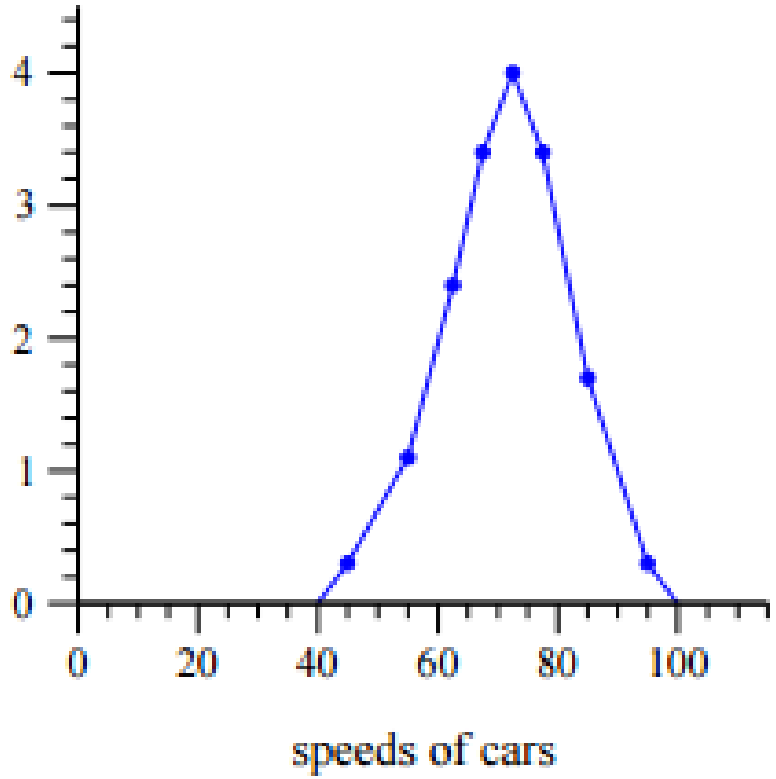

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Think about this: How do human social network degree distribution look like?

Normal distribution? Or other types of distribution?

Visualization: How do network degree distributions look like?

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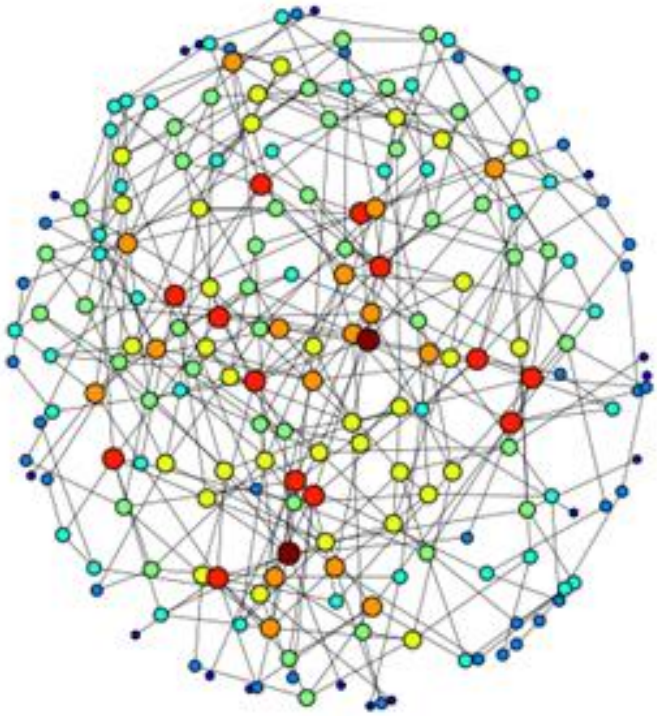

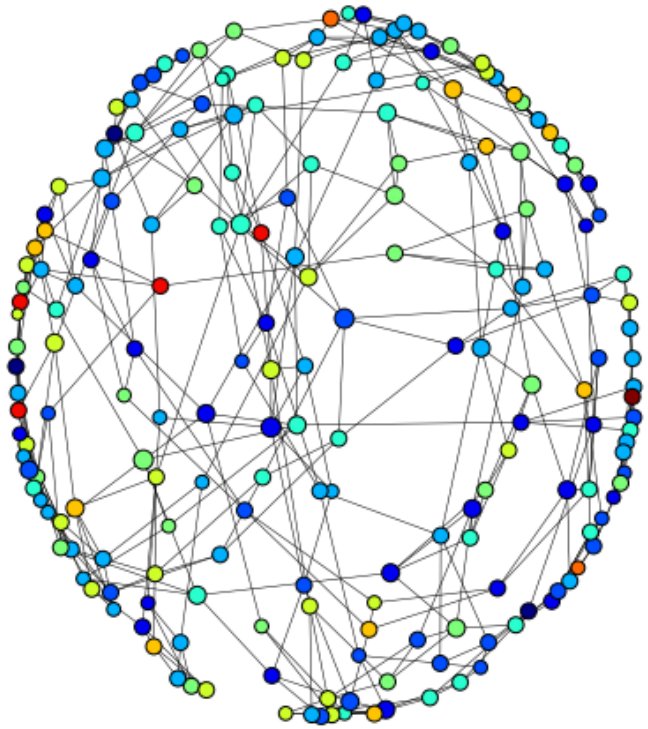

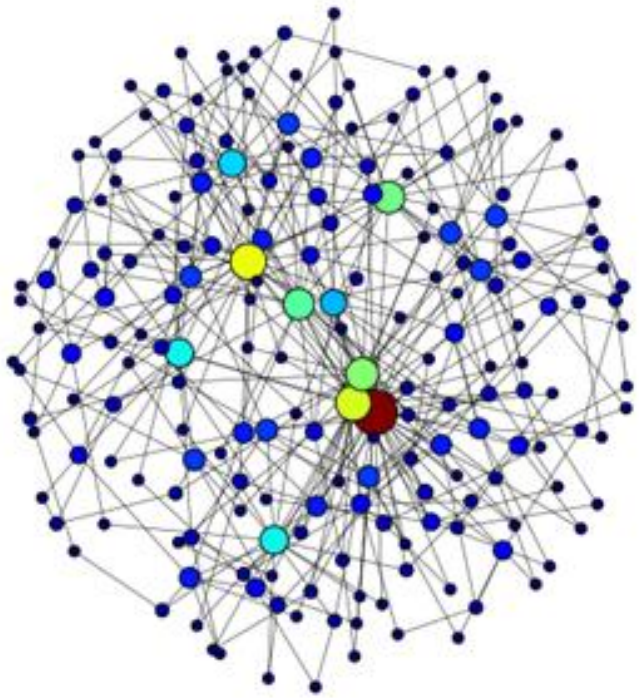

What are these three networks? How do their degree distribution look like?

Visualization: How do network degree distributions look like?

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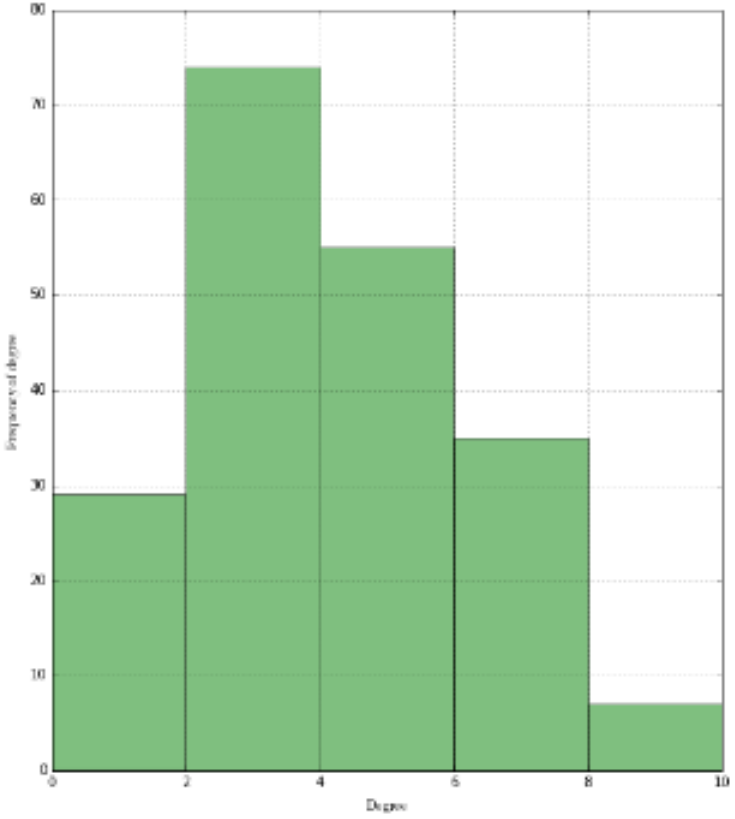
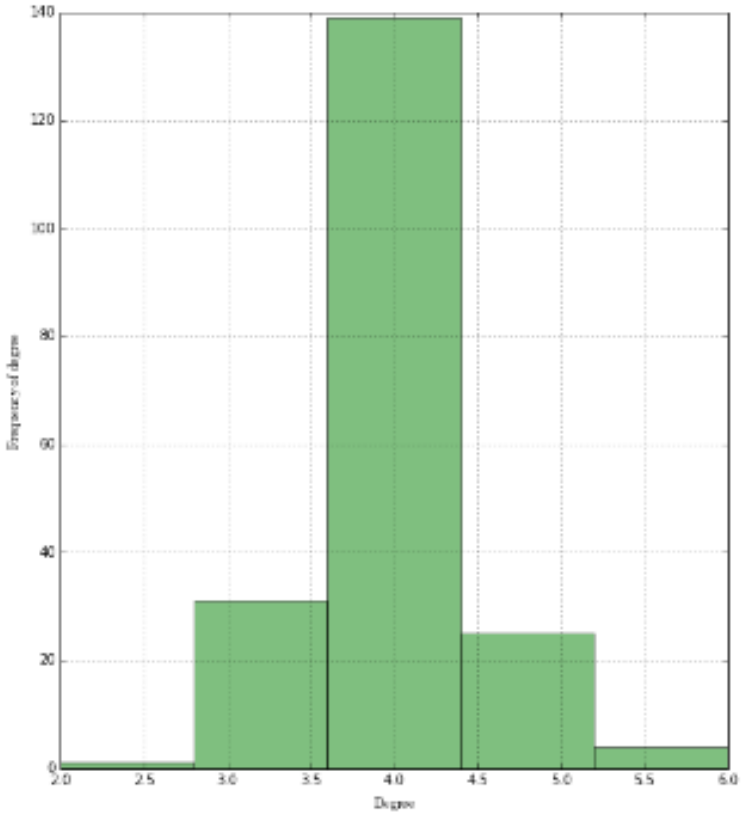
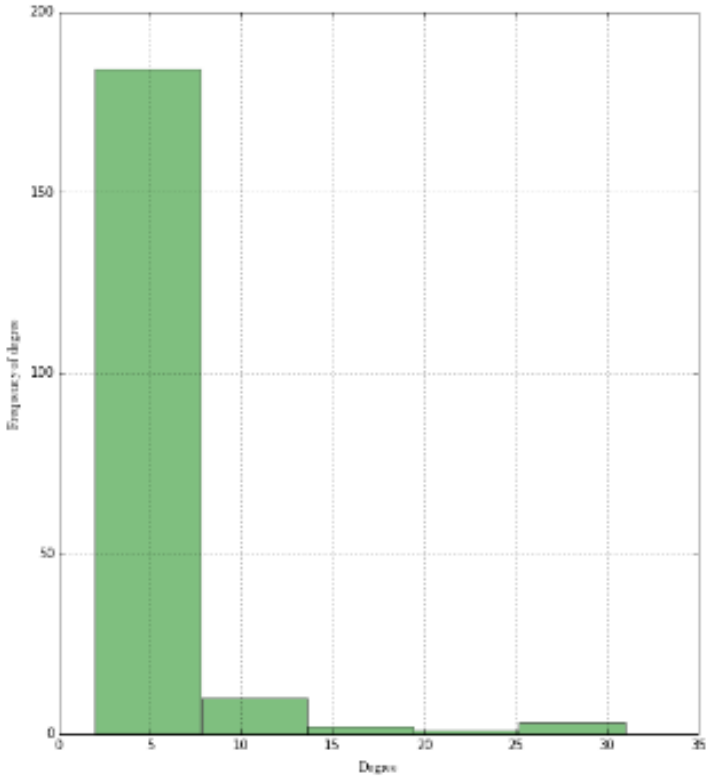


Random graph



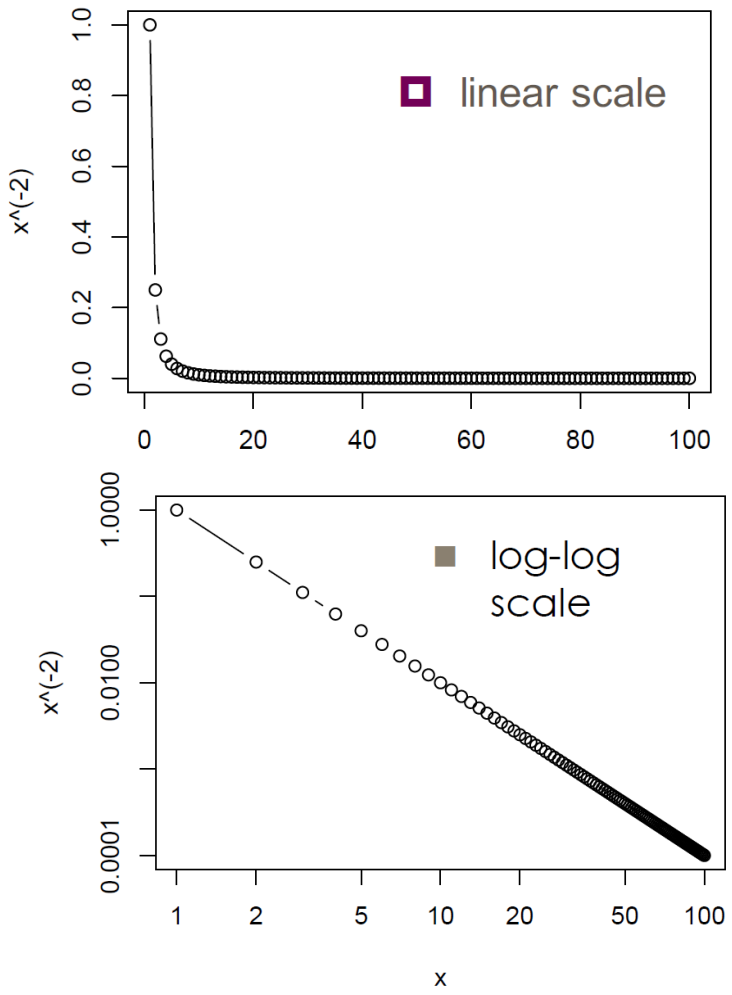
Small world graph



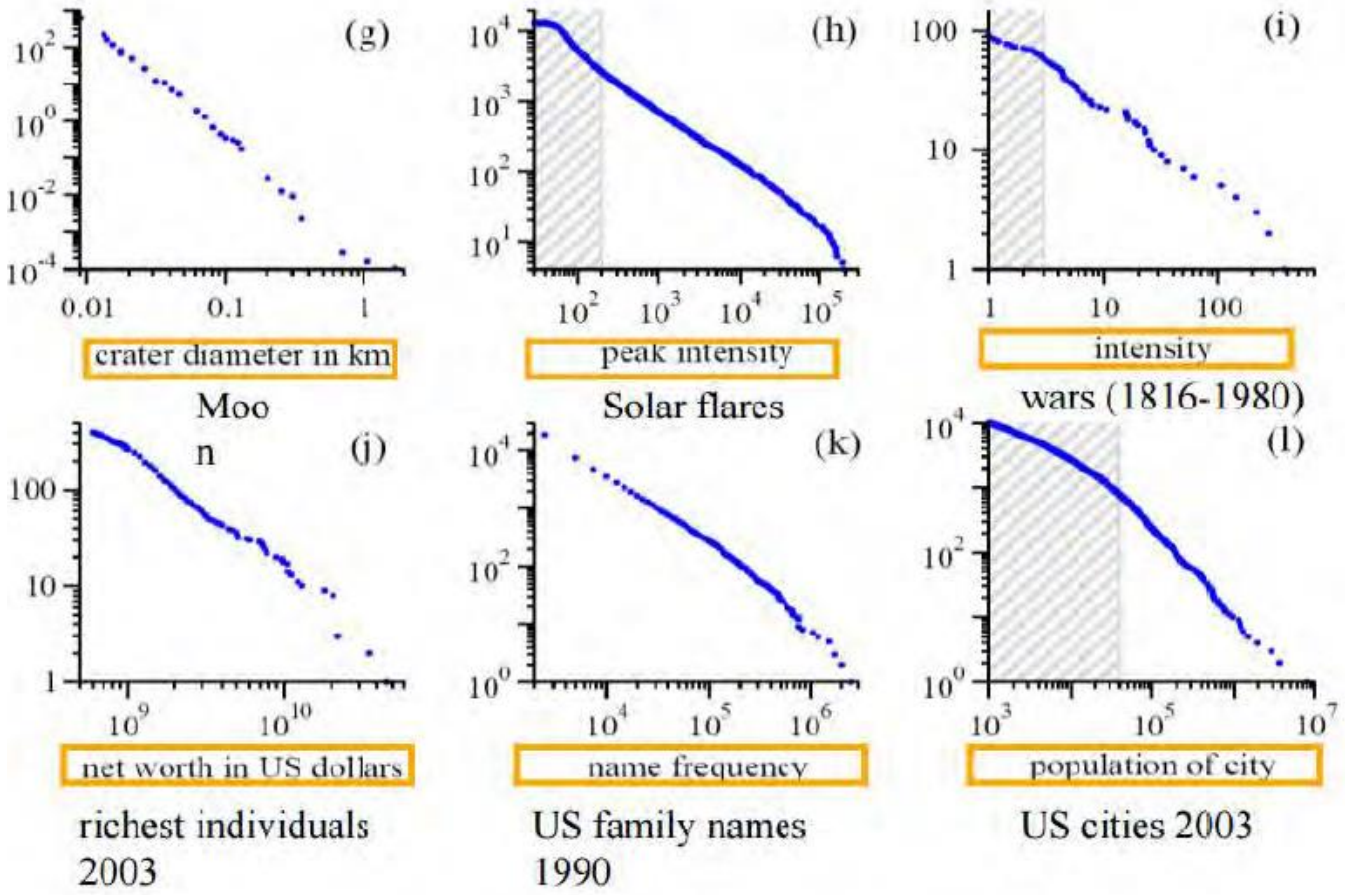
Scale free graph

Power law distribution

- Power – exponential, law – regularity
- High skew – asymmetric
- Heavy tailed – small values occurred highly frequently and large values highly infrequently
- Straight line on a log-log
- So what is the generic form of power-law distribution?
- And who follows power-law distribution?



Power-law Distribution
Real World examples



Source:MEJ Newman, ' Power laws, Pareto distributions and Zipf' s law', *Contemporary Physics* **46**, 323–351 (2005)

Power-law Distribution
Real World examples

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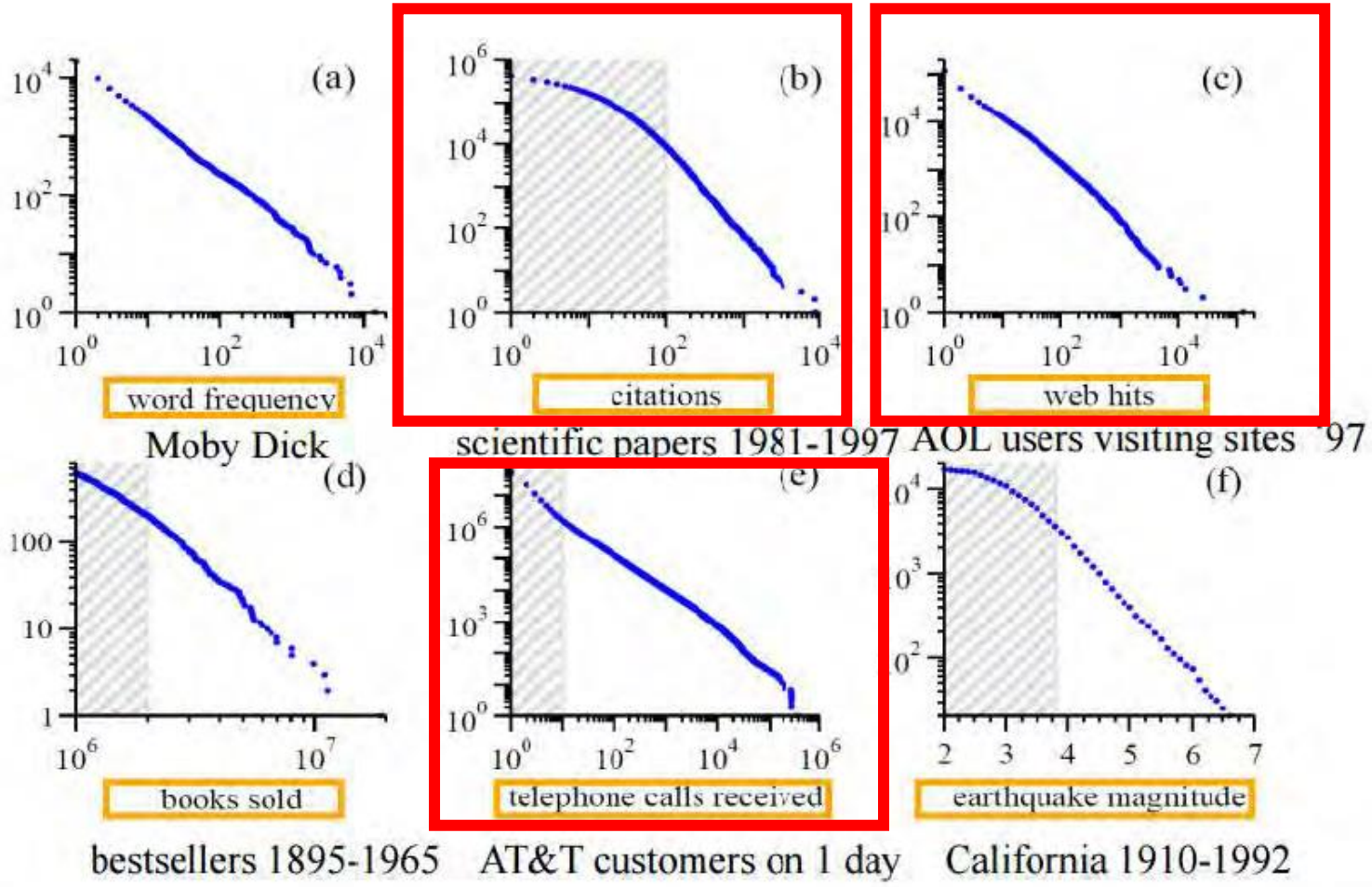

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Formulaic expression of power-law distribution

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- Straight line

$$p(x) = c + \alpha x$$

- Straight line on a log-log plot

$$\ln(p(x)) = c - \alpha \ln(x)$$

- And therefore --

$$p(x) = Cx^{-\alpha}$$

normalization constant (probabilities over all x must sum to 1)

power law exponent α

Power law distribution and scale-free

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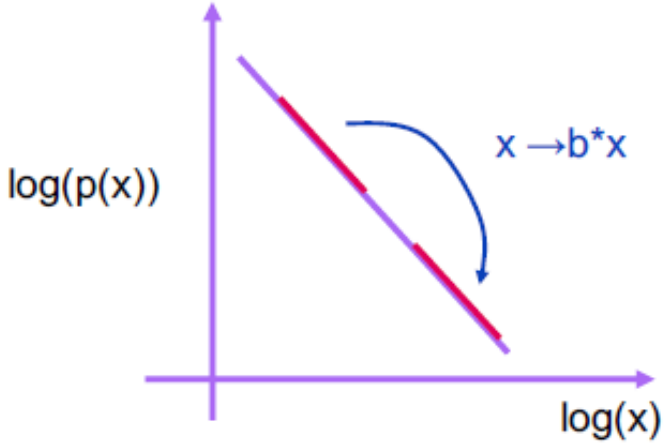
- A power law looks the same no matter what scale we look at it on (2 to 50 or 200 to 500)
- Only true of a power-law distribution!
- $p(bx) = g(b)p(x)$
- Scale-free definition: shape of the distribution is unchanged except for a multiplicative constant
- $p(bx) = c(bx)^{-\alpha} = c\, b^{-\alpha} x^{-\alpha} = c' x^{-\alpha}$



Networks with power-law features

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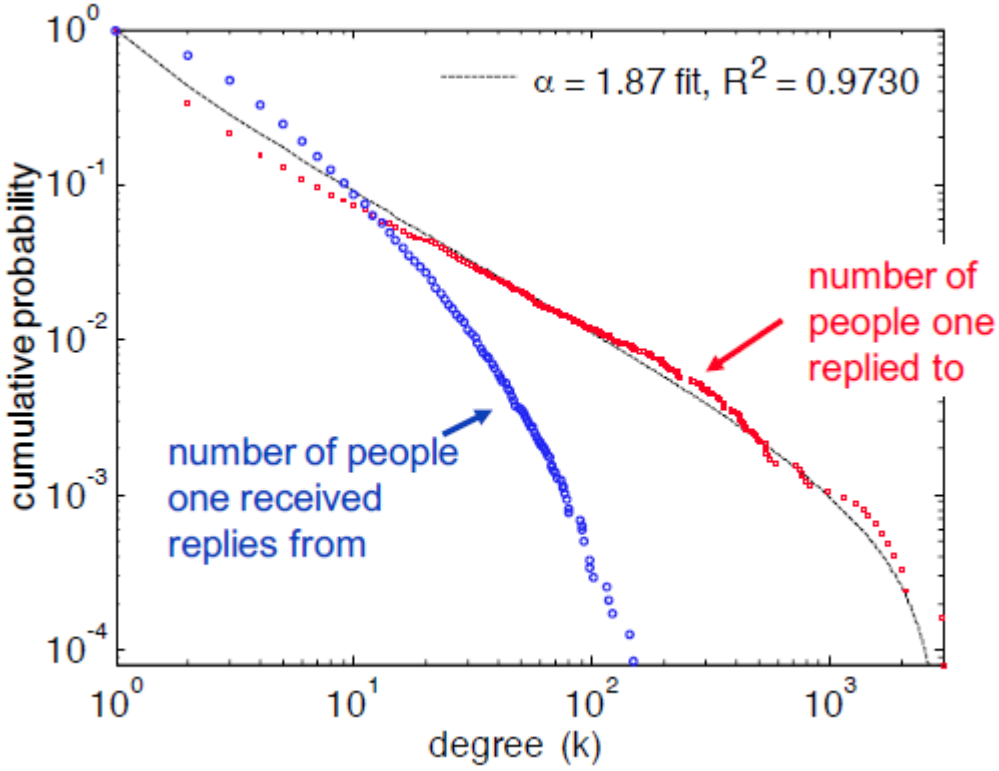
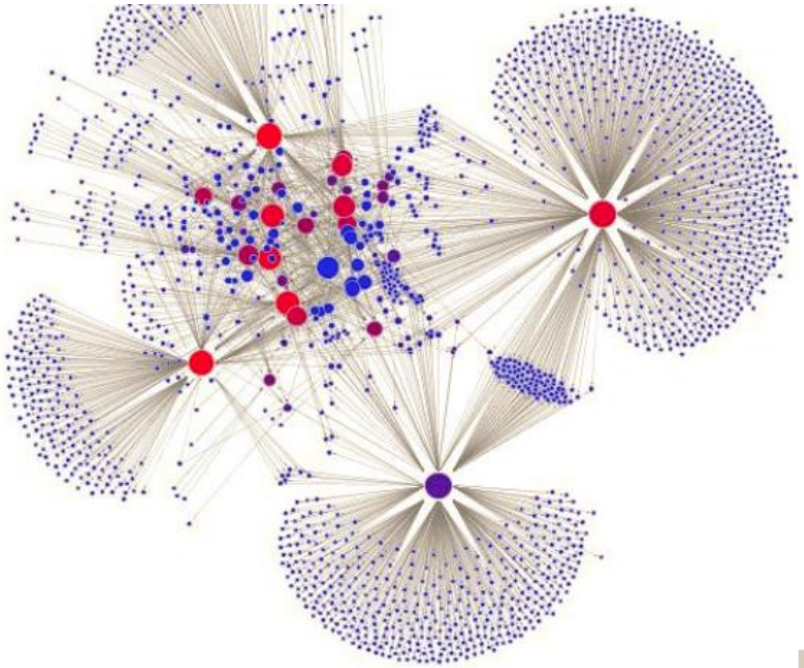
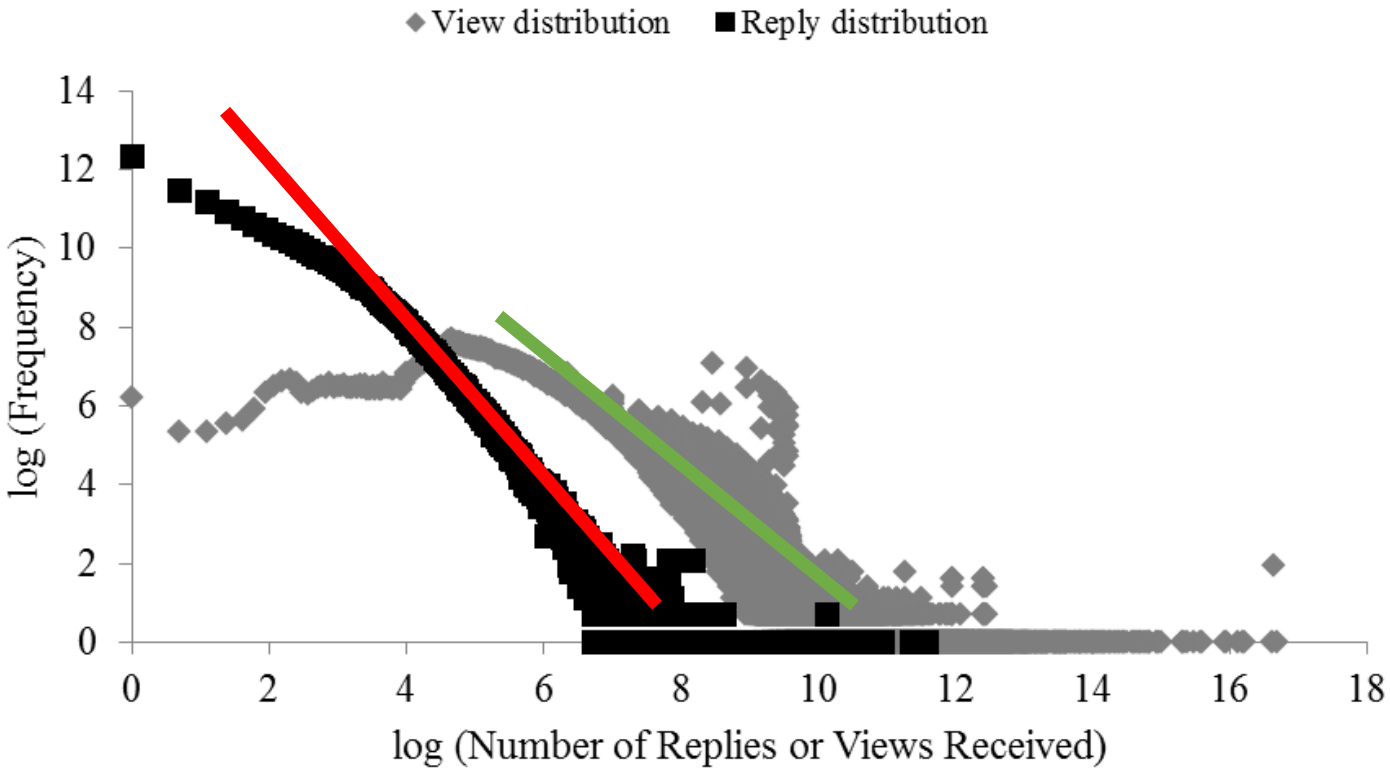
- Uneven participation: unit of analysis – individual participants





Networks with power-law features



Liang & Shen, 2013

- Frequency Distributions of View and Reply of All Discussion Threads
- Unit of analysis – individual posts

Bell curve vs. power law

What about random network?

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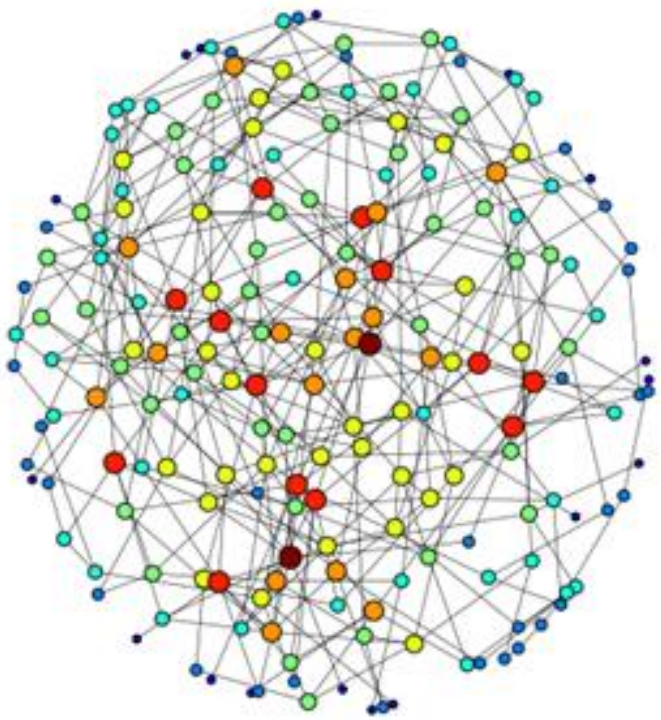

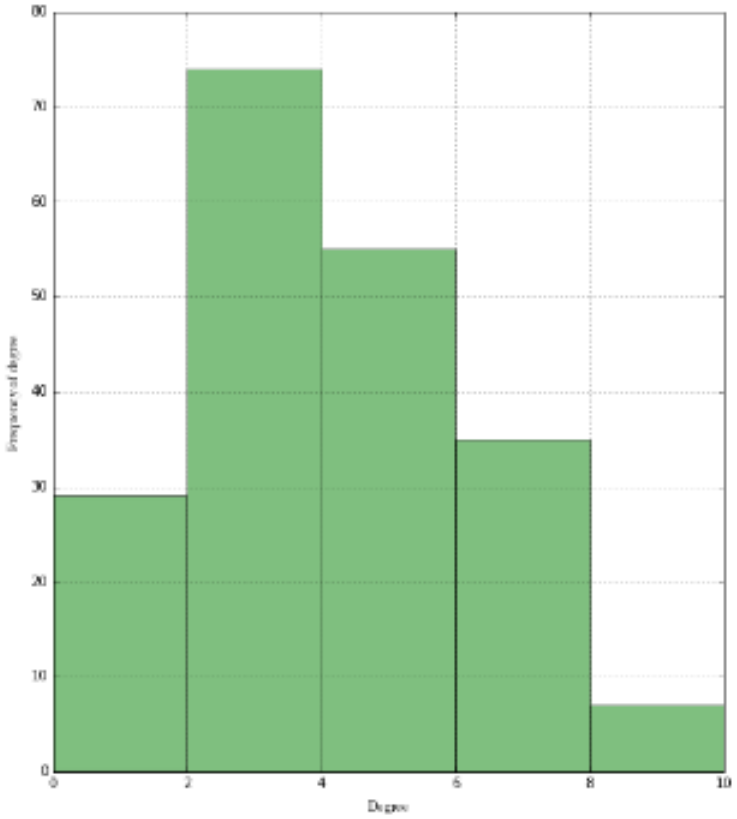

Erdos-Renyi graph: Poisson distribution

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- The binominal distribution gives us the probability that a node has degree k:

$$B(N-1;k;p) = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$

- The degree distribution of sufficiently large N random graph follows Poisson distribution – k being the number of edges, λ refers to average degree :

$$P(k \text{ events in interval}) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Poisson approximation

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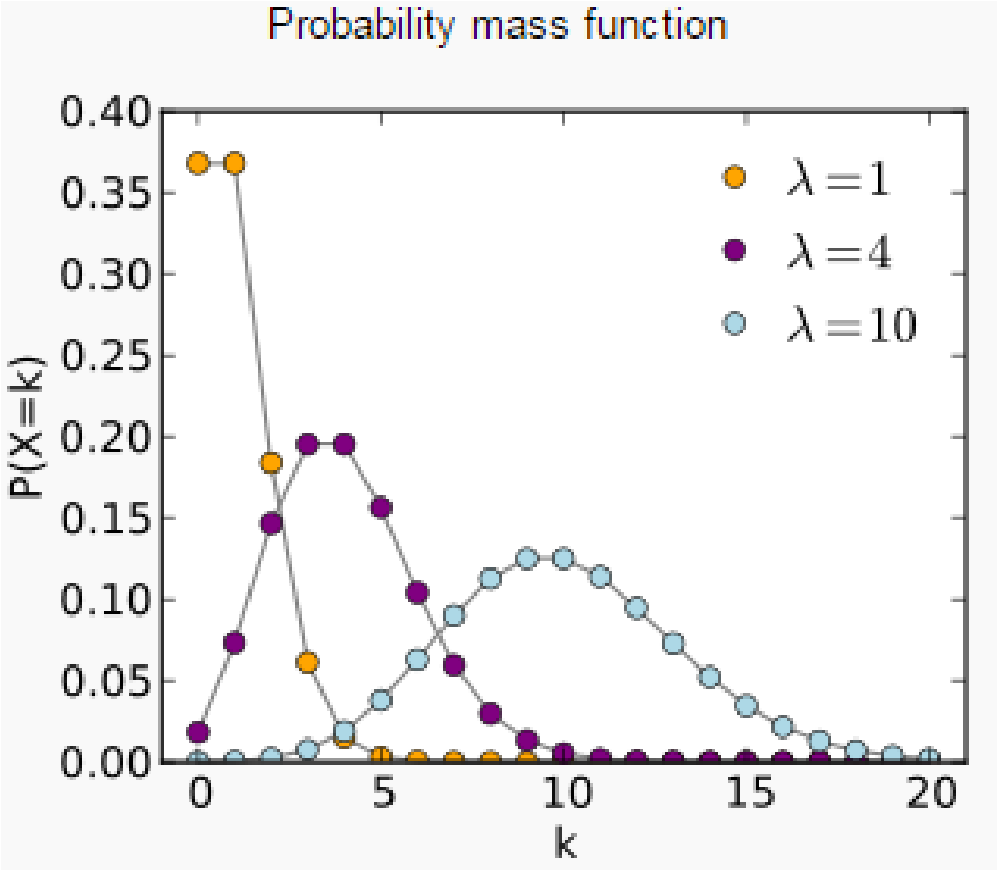
- The binomial distribution converges towards the Poisson distribution as the number of trials goes to infinity while the product np remains fixed or at least p tends to zero.
- $p(x = k) = \lim_{n\to\infty} \frac{n!}{k!(n-k)!} \left(\frac{\alpha}{n}\right)^k \left(1 - \frac{\alpha}{n}\right)^{n-k} = \frac{\alpha^k e^{-\alpha}}{k!}$

Binominal distribution

The pdf of Poisson distribution

Poisson distribution visualization



Why scale-free?

The mechanisms behind

Why scale-free?

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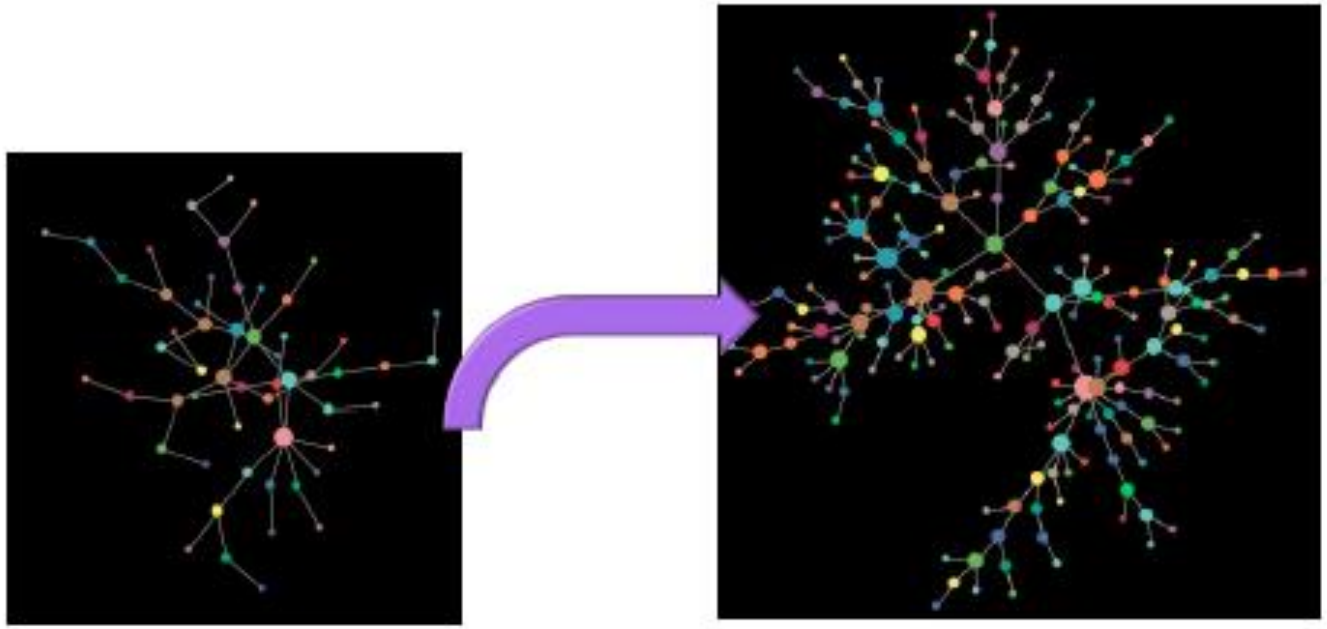
- Nodes grow over time – two mechanism generating a scale-free network: 1. older nodes got more connections; 2, larger nodes got more connections (preferential attachment)



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- Networks expand continuously by the addition of new vertices
- New vertices attach preferentially to existing vertices that are already well connected
- A model based on these two ingredients reproduces the observed scale-free distributions, which indicates that the development of large networks is governed by robust self-organizing phenomena that go beyond the particulars of individual systems. (Barabasi & Albert, 1999)

Random growth over time: age and degree

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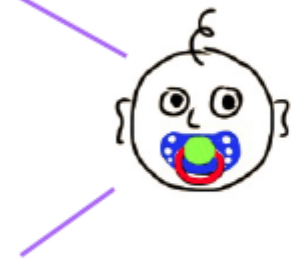
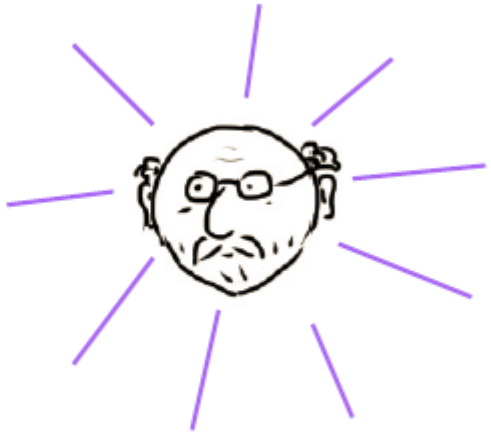
- Imagine the network generation mechanism as such:
  - Nodes appear one by one
  - Each new node will select m other nodes at random to connect to
  - Who will have higher chance of getting more connections?





Node degree growth speed

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- Older nodes get more degrees;
- In other words, the degree of node i, k_i is a function of time t.
- In order to know the degree of node i, it is important to start with knowing the speed of changing – the change in degree k_i (born at time i, 0< i < t)

$$\frac{dk_i(t)}{dt} = \frac{m}{t}$$

there are ***m*** new edges being added per unit time (with 1 new node)

the ***m*** edges are being distributed among ***t*** nodes

Node degree

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- How many new edges does a node accumulate since its birth at time i until time t?
- Integrate from t_i to t

$$\frac{dk_i(t)}{dt} = \frac{m}{t}$$

$$k_i(t) = m + m\log(\frac{t}{i})$$

born with ***m*** edges

A simplified example

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- m=1, i=1
- Total edges = 1+ ½+ 1/3 + ¼ + 1/5 …..
- $f(x) = \sum_1^n \frac{1}{n} \quad \sim \quad \int \frac{1}{x}$ = ln(x)

- Similarly, if m=m' and i=c
- Total edges = m+m*[1/c+1/(c+1)…..] = m+m*[f(x)-f(c)]= m+m*[ln(x)-ln(c)] = m+m*ln(x/c)

Preferential attachment

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- New nodes prefer to attach to well-connected nodes over less-well connected nodes
- Also known as
  - Cumulative advantage
  - Rich-get-richer
  - Matthew effect
- Mechanism
  - Visibility
  - Herding

Simulation of scale-free network

Barabasi-Albert model

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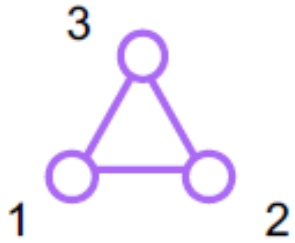
- First used to describe skewed degree distribution of the www
- Each node connects to other nodes with probability proportional to their degree
  - The process starts with some initial subgraph
  - Each new nodes comes in with m edges
  - Probability of connecting to node i

$$\Pi(i) = m \frac{k_i}{\sum_j k_j}$$

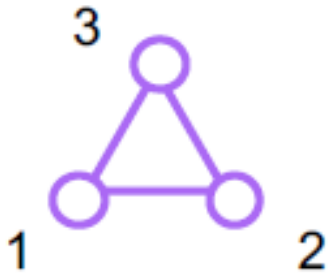
A simple BA model

- Supposing we start with an initial set of 3 fully connected nodes

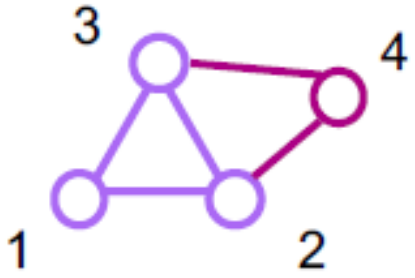


- Now add new nodes one by one, each one with m edges
- Each new edge connects to an existing nodes in proportion to the number of edges that node has – preferential attachment
-

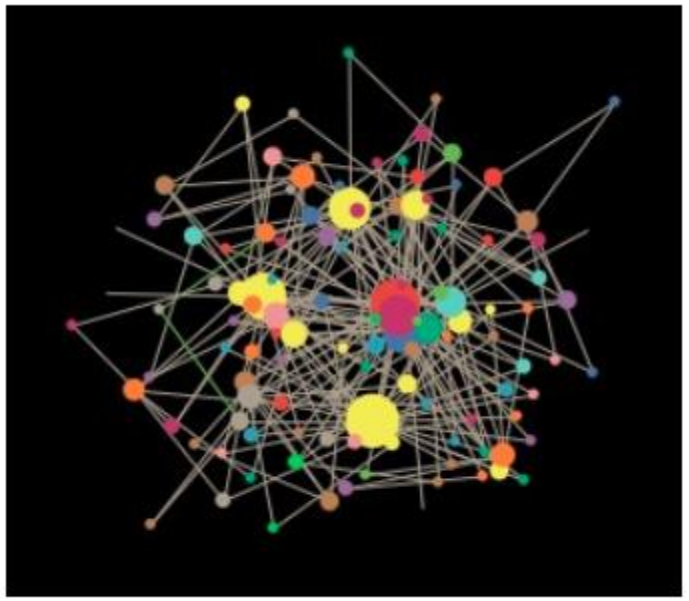
Generating a BA network
m=2

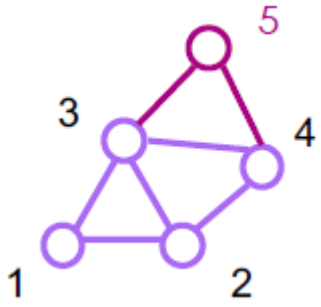


1 1 2 2 3 3



1 1 2 2 2 3 3 3 4 4





1 1 2 2 2 3 3 3 3 4 4 4 5 5

Random growth
vs. preferential growth

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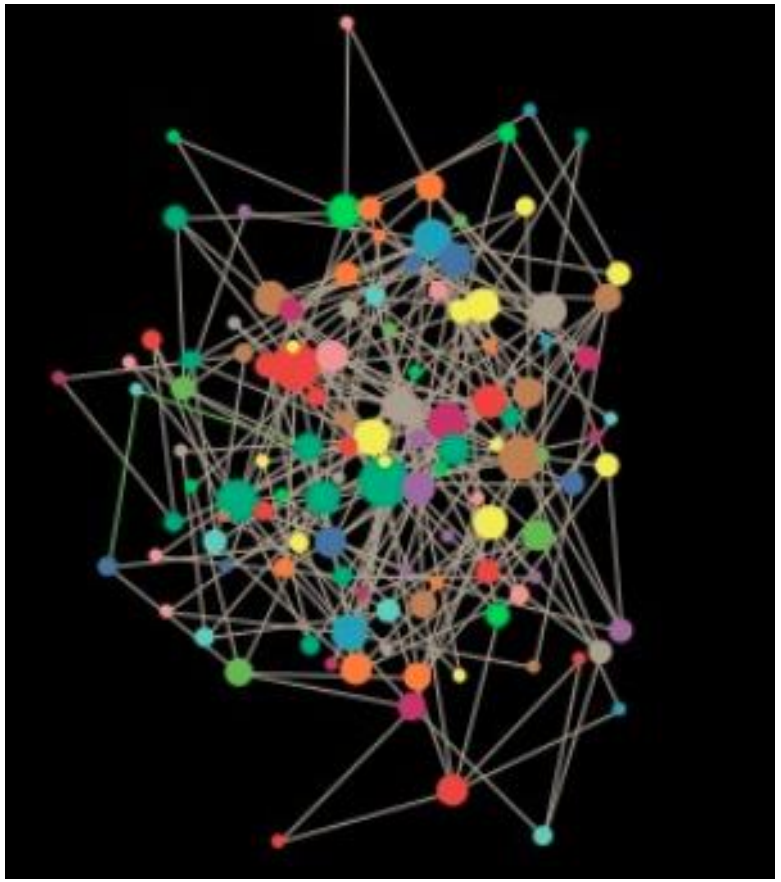

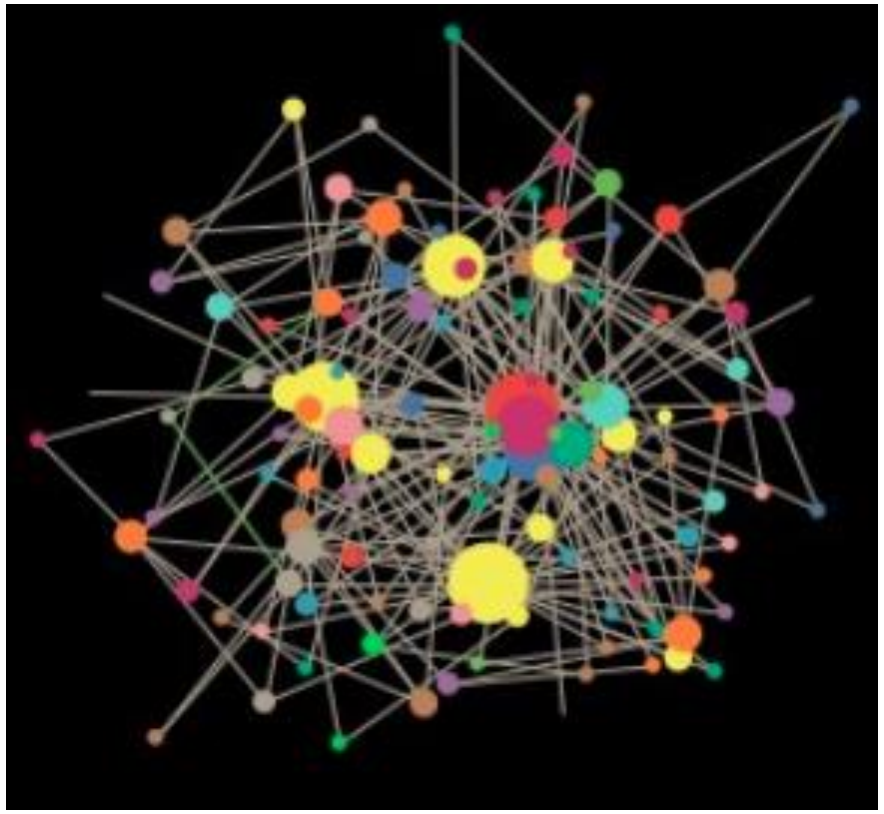

Not all scale-free networks are caused by preferential attachment

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- Other possible factors
- Homophily
  - Social selection
  - Social influence

Comparison

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- **Small world networks:**
- Observed what? Small average path length
- Why? Hybrid between local ties and random jumps (rewiring)
- Empirical evidence: social networks, co-authorship, citations, et.

- **Scale-free networks:**
- Observed what? Power-law distribution
- Why? Preferential attachment
- Empirical evidence: the internet, power grid, co-authorship, social media

Lab 4

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- Python basics for network analysis continued
- Next time: Quiz

Updated timetable

7	3/4	Lecture: Strength of weak ties Lab: Textual analysis example	Deadline for grouping – send to TA
8	3/11	Lecture: Community detection Lab: Gephi introduction Workshop 1: project proposal consultation	EK Chap 16, TK Chap 5
9	3/18	Lecture: Network data hypothesis testing Lab: UCINET introduction	EK Chap 19, TK Chap 6 Bring your topics and ideas to class
10	3/25	Workshop 2: Group project progress consultation	Bring your data set to class
11	4/1	**Reading week**	
12	4/8	Final test (Lecture 1 to Lecture 8) Workshop 3: Group project finding consultation	Bring your preliminary findings to class
13	4/15	**No Class**	
14	4/22	*Final Project Presentation Day*	Complete your project draft
	4/29		**Research paper due**

Quiz

- Close book, and close notes
- Format: multiple choice questions, short answer questions
- 90 minutes
- Covers L1-L4